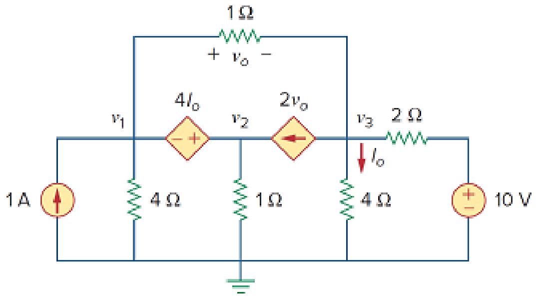
Chapter 3, Problem 31P

Problem

Find the node voltages for the circuit in Fig. 3.80.

Figure 3.80:



Step-by-step solution

Step 1 of 2

Refer to Figure 3.80 in the textbook.

There is a dependent voltage source present between the nodes with node voltages V_1 and V_2.

The two nodes are considered as a single node called the super node.

Write the Kirchhoff's Current Law equation at the super node.

$$-1+\frac{v_1-v_3}{1}+\frac{v_1}{4}+\frac{v_2}{1}+-2v_o=0$$

$$1+2v_o=\frac{v_1}{4}+\frac{v_2}{1}+\frac{v_1-v_3}{1} \quad \ldots\ldots (1)$$

But, $v_o=v_1-v_3$

Substitute v_1-v_3 for v_o in equation (1).

$$1+2(v_1-v_3)=\frac{v_1}{4}+\frac{v_2}{1}+\frac{v_1-v_3}{1}$$

$$1+2(v_1-v_3)=\frac{v_1+4v_2+4(v_1-v_3)}{4}$$

$$4+8(v_1-v_3)=v_1+4v_2+4(v_1-v_3)$$

$$4+8v_1-8v_3=v_1+4v_2+4v_1-4v_3$$

$$4+8v_1-8v_3=5v_1+4v_2-4v_3$$

$$4=-8v_1+8v_3+5v_1+4v_2-4v_3$$

$$4=-3v_1+4v_2+4v_3 \quad \ldots\ldots (2)$$

Step 2 of 2

Write the Kirchhoff's Current Law equation at the node 3, node labeled with node voltage v_3.

$$2v_O + \frac{v_3}{4} + \frac{v_3 - v_1}{1} + \frac{v_3 - 10}{2} = 0$$
$$8v_O + v_3 + 4(v_3 - v_1) + 2(v_3 - 10) = 0$$
$$8v_O + 7v_3 - 4v_1 - 20 = 0$$

Substitute $v_1 - v_3$ for v_O in the above equation.

$$8(v_1 - v_3) + 7v_3 - 4v_1 - 20 = 0$$
$$8v_1 - 8v_3 + 7v_3 - 4v_1 - 20 = 0$$

$4v_1 - v_3 = 20$ ...... (3)

Write the expression for the voltage v_2.

$v_2 = v_1 + 4I_O$, where $I_O = \frac{v_3}{4}$.

$$v_2 = v_1 + 4\left(\frac{v_3}{4}\right)$$
$$= v_1 + v_3$$

$v_1 - v_2 + v_3 = 0$ ...... (4)

Solve equations (2), (3) and (4) to obtain v_1, v_2 and v_3.

$$v_1 = 4.97 \text{ V}$$
$$v_2 = 4.8484 \text{ V}$$
$$v_3 = -0.12 \text{ V}$$

Thus, the required node voltages are, $\boxed{\begin{array}{l} v_1 = 4.97 \text{ V} \\ v_2 = 4.8484 \text{ V} \\ v_3 = -0.12 \text{ V} \end{array}}$.